ADVANCED MATHEMATICAL MODELS IN FINANCE
Final Project Part B

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1 Introduction

The product considered in this part of the project is a structured bond issued by Bank XX on the 31st of January 2023 with a four year maturity. The product is described through a swap agreement between Bank XX (Party A) and the investment bank IB (Party B).
Under the contract, Party A pays quarterly Euribor 3m plus a fixed spread of 300 basis points on a notional of €100 million. In exchange, party B pays an upfront equal to $X\%$ of the notional at the start date (2nd February 2023) and an equity linked coupon which is based on the performance of a basket composed by Enel and Axa stocks.
This coupon is determined through an Asian basket mechanism monitored annually over the lifetime of the contract. The basket performance is defined as the arithmetic average of the yearly returns of the two underlyings: ENEL and AXA. Specifically, the basket performance at maturity T from here onwards is called $S(t)$. The payoff at maturity is called $C(T)$.
The purpose of the project is to determine the fair upfront payment $X\%$ such that the initial value of the swap is zero under a single curve valuation framework.
In addition, the sensitivities of the structure with respect to the underlying assets and interest rates are computed in order to construct appropriate hedging strategies using the underlying stocks and an interest rate swap.

2 Contract Description

2.1 Swap Structure

The transaction considered in this project is a structured swap agreement between Bank XX (Party A) and the investment bank IB (Party B), with notional amount equal to €100 million and maturity equal to four years starting from the 2nd of February 2023.
Under the agreement:

- Party A pays quarterly Euribor 3m plus a fixed spread of 300 basis points;
- Party B pays an upfront amount equal to $X\%$ of the notional at inception;
- Party B also pays a structured equity-linked coupon at maturity.

The floating leg follows the Act/360 day count convention, while the structured coupon uses the 30/360 European convention.

2.2 Underlying Basket

The structured coupon depends on the performance of an equally weighted basket composed of stocks from Enel and Axa. The basket includes an Asian feature with yearly monitoring dates over the life of the contract. Let $(t_1, \ldots, t_4)$ denote the monitoring dates and let $(S_i(t_n))$ be the price of the i-th underlying observed at time t_n. The basket performance at maturity is defined as

$$S(T) = \frac{1}{2}\left(\frac{1}{4}\sum_{n=1}^{4}\frac{S^{(\text{Enel})}(t_n)}{S^{(\text{Enel})}(t_{n-1})}\right) + \frac{1}{2}\left(\frac{1}{4}\sum_{n=1}^{4}\frac{S^{(\text{Axa})}(t_n)}{S^{(\text{Axa})}(t_{n-1})}\right) \tag{1}$$

where: the factor 1/2 represents the equal weighting of the two assets, and the factor 1/4 corresponds to the arithmetic average over the four yearly monitoring dates.

2.3 Structured Coupon

At maturity, Party B pays a coupon linked to the positive performance of the basket. The payoff is given by

$$C(T) = \alpha \max\left(S(T) - 1, 0\right) \tag{2}$$

where the participation coefficient is fixed at $\alpha = 95\%$. Therefore, the investor participates in 95% of the positive basket performance above the initial level.

2.4 Definition of the Upfront Payment

The objective of the valuation is to determine the fair upfront payment $X\%$ such that the initial value of the swap is equal to zero under the adopted valuation framework.
The fair upfront payment is determined such that the initial net present value of all contractual cash flows is equal to zero. The analytical pricing decomposition is derived in Section 4.1.

3 Market and Modeling Framework

3.1 Market Setup

The valuation date is the 31st of January 2023 at 10:45 CET, which represents the observation point at which all market data are collected and the pricing is performed. Following the standard spot settlement convention for interest rate derivatives, the contract settlement date is the 2nd of February 2023, two business days after the valuation date, and coincides with the payment of the upfront amount $X\%$ by Party B. The swap has a maturity of four years, expiring on the 2nd of February 2027. The floating leg follows the Act/360 day count convention, consistent with the Euribor 3M market standard, with quarterly payment dates falling every three months from the settlement date. The equity-linked coupon is instead paid once at maturity. The four annual monitoring dates for the Asian basket fall on the 2nd of February 2024, 2025, 2026, and 2027, corresponding to the first through fourth anniversaries of the settlement date. Business day adjustments follow the Following convention with reference to the Eurex DE calendar, ensuring that all payment and monitoring dates falling on weekends or public holidays are rolled forward to the next valid business day within the same calendar month.

The notional amount of the swap is €100 million and remains constant throughout the life of the contract. Formally, the trade date is denoted t_0 = 31st January 2023, representing the date at which market data are observed and the contract is agreed upon, while the settlement date coincides with the start of the accrual period. The maturity of the contract is denoted $T = t_4$, corresponding to the 2nd of February 2027. The four annual monitoring dates are denoted (t_1, t_2, t_3, t_4), falling approximately one, two, three, and four years after the settlement date respectively, and represent the dates at which the stock prices of ENEL and AXA are recorded for the purpose of computing the Asian basket performance. All discounting is performed from the relevant cash flow date back to t_0, using the discount factors $D(t_0, t_i)$ extracted from the bootstrapped curve.

3.2 Interest Rate Representation and Bootstrap

The interest rate framework relies on three equivalent representations of the term structure: discount factors, continuously compounded zero rates, and forward rates. Given a discount factor $D(t_0, t_i)$, representing the present value at t_0 of one unit of currency paid at t_i, the associated continuously compounded zero rate is defined as

$$r(t_i) = -\frac{\ln D(t_0, t_i)}{\delta(t_0, t_i)} \tag{3}$$

where $\delta(t_0, t_i)$ denotes the year fraction between t_0 and t_i under the ACT/365 Fixed convention. The two representations are related by

$$D(t_0, t_i) = \exp(-r(t_i)\,\delta(t_0, t_i)) \tag{4}$$

For dates not coinciding with a bootstrapped pillar, the discount factor is obtained by linearly interpolating the continuously compounded zero rates between the two nearest pillars. The continuously compounded instantaneous forward rate between two consecutive dates is extracted as

$$f(t_{i-1}, t_i) = -\frac{\ln D(t_0, t_i) - \ln D(t_0, t_{i-1})}{\delta(t_{i-1}, t_i)} \tag{5}$$

and is used as the risk-neutral drift in the Monte Carlo simulation.
The bootstrap procedure constructs the discount curve sequentially from market quotations of deposits, futures, and swaps, enforcing the no-arbitrage condition that each instrument is priced exactly at par. The curve extends to February 2027, yielding a discount factor from valuation date to settlement of $D(t_0, t_{\text{set}}) = 0.999894$ and a four-year discount factor of $D(t_0, T) = 0.892192$.

3.3 GBM model

In order to model the dynamics of the underlying asset price, we assume that the assets follow a Geometric Brownian Motion (GBM).
Under the risk-neutral probability measure, the asset price process $(S_t)_{t\geq 0}$ satisfies the Stochastic Differential Equation

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t \tag{6}$$

where r is the risk free interest rate under continuous compounding, q is the continuous dividend yield, σ is the constant volatility, W_t is a standard Brownian motion under the same risk neutral measure.
This model is used as a simulation engine to generate risk neutral paths of the underlying assets. Since the payoff of the product considered in this work is path dependent, pricing is carried out numerically via Monte Carlo simulation rather than closed form valuation. The Monte Carlo implementation is discussed in section 4.5.
Since the setting is multivariate the model is extended to a system of correlated GBM processes. In particular, the Brownian Motions are not independent, but exhibit constant correlation ρ.
This is implemented by defining a correlation matrix:

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \tag{7}$$

and sampling the driving shocks (Z_1, Z_2) at each time step as a correlated bivariate normal vector with covariance structure Σ. This corresponds to the continuous time correlation structure:

$$dW_t^{(1)} dW_t^{(2)} = \rho\, dt, \tag{8}$$

and preserves the GBM dynamic of the individual asset.

Parameter	Description	ENEL	AXA
S_0	Initial spot price (€)	100	200
σ	Annualised volatility	0.162	0.2
d	Dividend yields	0.025	0.029
ρ	Equity correlation	0.4	

Table 1: Market parameters used in the Monte Carlo simulation, observed at the valuation date 31st January 2023.

4 Pricing methodology and implementation

4.1 Pricing Decomposition

At inception (31st January 2023, 10:45 CET), the fair value of the product is obtained under the described framework as

$$X_{\text{upfront}} = PV_{\text{Euribor}} + PV_{\text{SPOL}} - PV_{\text{Coupon}}. \tag{9}$$

The bond is priced consistently with the swap termsheet, where Party A pays Euribor 3M + spread, and Party B pays an upfront amount and an equity-linked coupon at maturity.

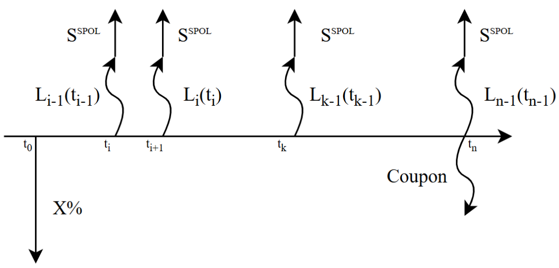

Figure 1: Cash flow involved from the IB perspective.

4.2 Euribor floating leg

The present value of the floating Euribor leg over the 4-year horizon telescopes. Mathematically, the sum of the discounted forward rates collapses into a simple difference:

$$PV_{\text{Euribor}} = \sum_{i=1}^{16} L(t_{i-1}, t_i) \cdot \delta_i \cdot B(0, t_i) = 1 - B(0, T)$$

where $T = 4$ years. In terms of implementation, there is no need to iteratively calculate and sum the individual quarterly forward rates; the value is obtained directly in a single step by subtracting the final 4-year discount factor from 1.

4.3 Spread over libor floating leg (SPOL)

The present value of the spread leg is straightforward, as it consists of fixed deterministic payments. Mathematically, it is the sum of the fixed spread s multiplied by the respective year fraction δ_i and discounted to today:

$$PV_{\text{Spread}} = s \cdot \sum_{i=1}^{16} \delta_i \cdot B(0, t_i)$$

In the implementation, this is achieved by computing the sum of the product of the fixed 300 bps spread, the exact day-count fractions, and the corresponding discount factors across all 16 quarterly payment dates.

4.4 Coupon fixed leg

The structured coupon represents the only stochastic component of the transaction and is linked to the positive performance of the Asian equity basket defined in Equation 1. The payoff paid by Party B at maturity is

$$C(T) = \alpha \max(S(T) - 1, 0) \tag{10}$$

where $\alpha = 95\%$ is the participation coefficient and $S(T)$ is the Asian basket performance observed over the four annual monitoring dates.
Since the payoff depends on the entire path of the underlying assets through an average, no closed form pricing formula is available under the adopted multivariate GBM framework. The present value of the coupon leg is therefore computed numerically using Monte Carlo simulation under the risk neutral measure.

$$PV_{\text{Coupon}} = B(0, T)\, \mathbb{E}^{\mathbb{Q}} \left[\alpha \max(S(T) - 1, 0)\right] \tag{11}$$

where $B(0, T)$ is the discount factor extracted from the bootstrapped curve and $\mathbb{Q}$ denotes the risk neutral probability measure.
In the implementation, correlated paths for ENEL and AXA are simulated daily under the Geometric Brownian Motion dynamics described in Section 3.3. For each simulated path, the basket performance is computed according to Equation 1, the terminal payoff is evaluated, and the discounted expectation is estimated as the average across all Monte Carlo scenarios.
Formally, given N_{sim} simulated scenarios, the Monte Carlo estimator of the coupon present value is

$$\widehat{PV}_{\text{Coupon}} = B(0, T)\, \frac{1}{N_{\text{sim}}} \sum_{k=1}^{N_{\text{sim}}} \alpha \max\left(S^{(k)}(T) - 1, 0\right) \tag{12}$$

where $S^{(k)}(T)$ denotes the basket performance computed along the k-th simulated path.

4.5 MC Implementation

The valuation of the structured coupon is carried out using Monte Carlo framework under the risk neutral measure. The simulation is based no the multivariate Geometric Brownian motion dynamics described in Section 3.3, with constant volatilities, deterministic dividend yields, and a constat correlation structure between the underlying ENEL and AXA stocks.

The time discretization is performed on a daily grid with 254 days per year, resulting in 1016 simulation steps over the four year time framework.
At each timestep, correlated standard normal shocks are generated through a bivariate normal distribution with covariance matrix reflecting the instantaneous equity correlation.
The asset paths are updated recursively using the discretization introduced by GBM:

$$S_{t+\Delta t} = S_t \exp\left[\left(r - q - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_t\right]. \tag{13}$$

Although the simulation is performed at a daily frequency to preserve the stochastic evolution of the underlying processes, only the prices corresponding to the four annual monitoring dates are retained. These observations are then used to compute the arithmetic returns of each underlying and to construct the Asian basket performance.
For each simulated trajectory, the basket payoff is evaluated at maturity and discounted back to the valuation date using the bootstrapped discount factor $B(0,T)$. The Monte Carlo estimator of the coupon leg is obtained as the empirical average over all simulated scenarios as described in Equation 12.
The accuracy of the estimator is controlled through the number of simulated paths, while consistency of the implementation is validated via the geometric basket benchmark discussed in Section 7.1.

4.6 Pricing Results

The valuation framework described above yields the following decomposition of the fair upfront payment expressed as a fraction of notional.

Component	Present Value
Floating Euribor Leg	0.1078
Spread Leg	0.1143
Equity Coupon Leg	0.0262
Fair Upfront X	0.1959

Table 2: Pricing decomposition expressed as fraction of notional

The fair upfront payment is therefore equal to 19.59% of the notional. The deterministic interest-rate components account for the majority of the product value, while the structured equity-linked coupon contributes approximately 2.62% of notional.

5 Greeks and Sensitivity Analysis

5.1 Equity Deltas

The equity sensitivities of the upfront payment are computed via a bump-and-revalue approach: the initial price of each underlying is increased by one unit (€1), the entire product is repriced via Monte Carlo on the same random paths, and the delta is approximated as the finite difference:

$$\Delta_i = \frac{X\%\big(S_0^{(i)} + 1\big) - X\%\big(S_0^{(i)}\big)}{€1} \tag{14}$$

5.2 Interest Rate DV01

A parallel shift of the entire bootstrapped discount curve by +1 basis point increases the upfront from 19.58% to 19.61%, yielding

$$DV01 = X\%_{\text{bumped}} - X\%_{\text{base}} = +0.03\% \text{ of notional} = +€\,30{,}000 \tag{15}$$

The positive sign reflects the structure of the product. Since the present value of the telescoping Euribor leg is $1 - B(0,T)$, an increase in interest rates reduces the discount factor at maturity $B(0,T)$, leading to a direct increase in the Euribor leg's value. Conversely, the spread leg behaves as a series of fixed deterministic cash flows and loses value when discounted at higher rates. The gain in the Euribor leg dominates the loss in the spread leg, causing the net required upfront to increase with rising rates and resulting in a positive DV01.

6 Hedging Strategy

6.1 Delta Hedge via ENEL and AXA Shares

Executing the bump-and-revalue procedure described in Section 5.1, the obtained sensitivities are

$$\Delta_{\text{Enel}} = 0, \qquad \Delta_{\text{Axa}} = 0$$

This is not a numerical artifact but an exact analytical property of the payoff structure.
Under the GBM model, the only equity-sensitive component of the upfront is PV_{Coupon}, whose value depends on the basket performance $S(T)$, defined as an arithmetic average of sequential annual returns. For each underlying s, the stock price can be written as

$$S_t^{(s)} = S_0^{(s)} \exp\left[\left(r - q_s - \frac{\sigma_s^2}{2}\right)t + \sigma_s W_t\right] \tag{16}$$

Therefore, each return ratio entering the basket is

$$\frac{S^{(s)}(t_n)}{S^{(s)}(t_{n-1})} = \frac{S_0^{(s)} X^{(s)}(t_n)}{S_0^{(s)} X^{(s)}(t_{n-1})} = \frac{X^{(s)}(t_n)}{X^{(s)}(t_{n-1})} \tag{17}$$

where $X^{(s)}(t)$ denotes the stochastic exponential term. The initial spot $S_0^{(s)}$ cancels exactly from every ratio, implying that the basket performance $S(T)$ is independent of the initial stock prices. As a consequence, the coupon payoff and its present value are also independent of $S_0^{(\text{Enel})}$ and $S_0^{(\text{Axa})}$. Hence, bumping either stock price by €1 leaves the product value unchanged,

$$PV\left(S_0^{(s)} + 1\right) = PV\left(S_0^{(s)}\right) \tag{18}$$

which implies

$$\Delta_{\text{Enel}} = \Delta_{\text{Axa}} = 0 \tag{19}$$

This result holds exactly and is therefore independent of the bump size and of the number of Monte Carlo simulations. Consequently, no equity hedge is required for this product.

6.2 Interest Rate Hedge via a 4-Year IRS

The interest rate risk is hedged by entering an offsetting interest rate swap. Since the product has a positive DV01 (the upfront increases when rates rise), the hedge requires a position that loses value when rates rise, namely a **receiver IRS** in which the hedger pays Euribor 3M and receives fixed. This position gains value when rates fall and loses value when rates rise, exactly offsetting the rate exposure of the structured bond.
The hedge notional is determined by matching the DV01 of the product against the DV01 of the 4-year IRS per unit notional:

$$N_{\text{IRS}} = -\frac{DV01_{\text{product}}}{DV01_{\text{IRS, unit notional}}} \tag{20}$$

The numerical result gives a receiver IRS notional of approximately **€787,593 per €1,000,000 of product notional**, or equivalently €78.76 million on the full €100 million trade.
The DV01 of the structured product is smaller than that of a plain vanilla IRS because the equity-linked coupon is not interest-rate sensitive, and the positive rate exposure of the telescoping Euribor leg is partially offset by the negative rate exposure of the spread leg. The residual rate risk comes almost entirely from the spread leg, which behaves as a series of fixed cash flows.
After applying both hedges, the portfolio presents no residual equity delta and no parallel DV01, satisfying the hedging requirements of the assignment.

7 Extension of Project Requests

7.1 Validation of MC

As a consistency check for the Montecarlo Princing engine, we consider a geometric basket payoff for which an exact closed form solution is available. Since both instruments are driven by the same underlying

Geometric Brownian Motion dynamics and correlation structure, agreement between MC and closed form prices validates the correctness of the simulation paths, drifts, and dependency structure.
This benchmark does not validate the arithmetic Asian payoff but enables a reliability check of the numerical framework used for its valuation.
We define the geometric basket payoff

$$G = \sqrt{\left(\prod_{n=1}^{N} R_{\text{Enel}}^{(n)}\right)^{1/N} \left(\prod_{n=1}^{N} R_{\text{Axa}}^{(n)}\right)^{1/N}} \tag{21}$$

The corresponding discounted payoff is

$$V_G = D(0,T)\,\alpha\,\mathbb{E}\left[(G-1)^+\right] \tag{22}$$

Given the risk neutral measure, each asset follows a Geometric Brownian Motion,

$$\frac{dS_t}{S_t} = (r-q)\,dt + \sigma\,dW_t \tag{23}$$

Solving the Stochastic Differential Equation gives

$$S_T = S_0 \exp\left(\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma W_T\right) \tag{24}$$

Implying that S_T is log normally distributed.
Each return ratio $R_s^{(n)}$ is therefore also lognormal. Since logarithms transform products into sums, we obtain

$$\log G = \frac{1}{2N} \sum_{n=1}^{N} \left(\log R_{\text{Enel}}^{(n)} + \log R_{\text{Axa}}^{(n)}\right). \tag{25}$$

which is a linear combination of Gaussian variables, thus $log\,G$ is normally distributed. So the payoff $(G-1)^+$ corresponds to an European call option on a lognormal underlying with strike $K = 1$, and can be priced in closed form using the Black–Scholes formula.
Both the arithmetic and geometric baskets are evaluated on the same Monte Carlo paths, ensuring that any discrepancy between the two is solely due to the payoff structure and not to simulation noise.
This allows a direct comparison between Monte carlo estimates and the analytical benchmark. Observing consistency between Monte Carlo and closed-form prices provides evidence of correct implementation of:

- The risk neutral drift adjustment
- the correlation structure of the underlying Brownian motions
- the consistency of the discounting simulation framework

The observed ordering between arithmetic and geometric baskets is consistent with Jensen's inequality, as expected from the concavity of the geometric aggregation.

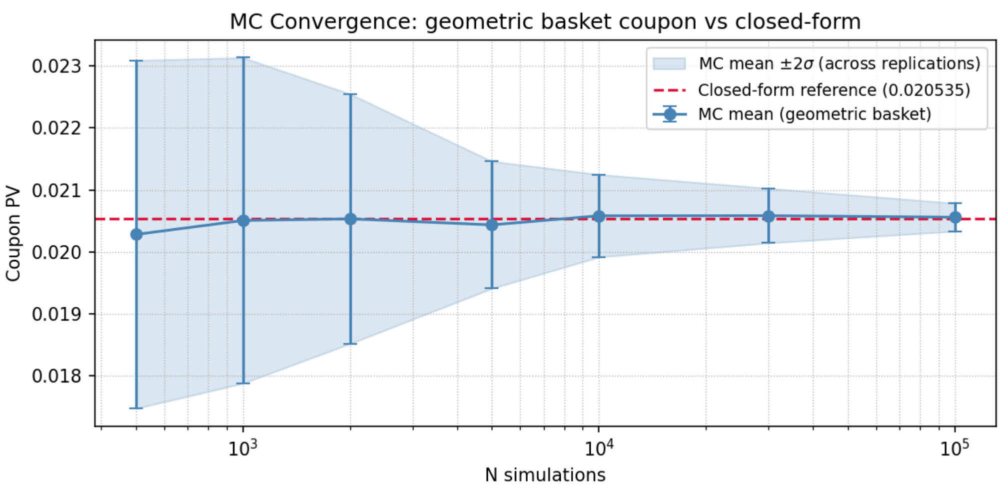

Figure 2: Monte Carlo convergence of the geometric basket price toward the closed-form benchmark

The small numerical discrepancy between Monte Carlo and closed-form prices is fully consistent with sampling error and confirms the stability of the implementation.

8 Results

The fundamental conclusion of this valuation is that the structured swap behaves predominantly as an interest rate-driven instrument rather than an equity-linked one. The fair upfront payment is almost entirely determined by the combination of the Euribor floating leg and the fixed spread leg, while the optional component linked to the equity basket exerts only a minimal impact on the final value of the structure.

In terms of risk management, the equity Deltas for both Enel and Axa are exactly zero, representing an exact analytical property of the payoff rather than a numerical artifact. Because the structured coupon is based on an Asian performance calculated as an arithmetic average of sequential period returns, the initial spot levels mathematically cancel out within the price ratios. Consequently, the entire payoff distribution remains completely independent of the initial stock values, meaning that the product requires no directional equity hedging strategy at inception.

Conversely, the structure exhibits a clear exposure to interest rate risk, with the upfront value showing a positive sensitivity to yield curve shifts. To completely neutralize this interest rate exposure, the optimal hedging strategy requires entering a mirror receiver Interest Rate Swap, with its notional precisely calibrated to offset the rate sensitivity of the original structured bond.

Finally, the accuracy and robustness of the daily Monte Carlo simulation engine were rigorously validated by pairing the model with an analytical benchmark based on a geometric basket evaluated over the exact same stochastic paths. The precise convergence of the simulated geometric price toward the Black–Scholes closed-form solution confirms the exact implementation of the risk-neutral drift and the underlying correlation structure. Furthermore, the systematic positive gap observed between the arithmetic and geometric basket prices fully respects Jensen's inequality, confirming the overall numerical stability and theoretical consistency of the framework.

9 Appendix

9.1 Pseudocode

Algorithm 1 Discount Curve Bootstrap

Load deposits, futures, swaps, settlement date t_0 from CSV
$\mathcal{D} \leftarrow \{(t_0, 1.0)\}$ ▷ initialise with $B(t_0, t_0) = 1$
for each deposit maturity $t_i \le t_{\text{fut}}^{(1)}$ **do**
  $B(t_0, t_i) \leftarrow \dfrac{1}{1 + L_i^{\text{mid}} \cdot \delta_i^{\text{ACT/360}}}$
  $\mathcal{D} \leftarrow \mathcal{D} \cup \{(t_i,\, B(t_0, t_i))\}$
end for
for each future $k = 1, \ldots, 7$ **do**
  $B^{\text{interp}}(t_0, t_s^{(k)}) \leftarrow \textsc{LinInterpZeroRate}(\mathcal{D},\, t_s^{(k)})$
  $F(t_s^{(k)}, t_e^{(k)}) \leftarrow \dfrac{1}{1 + f_k^{\text{mid}} \cdot \delta_k^{\text{ACT/360}}}$
  $B(t_0, t_e^{(k)}) \leftarrow F(t_s^{(k)}, t_e^{(k)}) \cdot B^{\text{interp}}(t_0, t_s^{(k)})$
  $\mathcal{D} \leftarrow \mathcal{D} \cup \{(t_e^{(k)},\, B(t_0, t_e^{(k)}))\}$
end for
$\text{BPV} \leftarrow \delta(t_0, t_1^{\text{sw}})^{30E/360} \cdot B^{\text{interp}}(t_0, t_1^{\text{sw}})$
for each swap maturity t_n^{sw}, rate s_n **do**
  $B(t_0, t_n^{\text{sw}}) \leftarrow \dfrac{1 - s_n \cdot \text{BPV}}{1 + s_n \cdot \delta_n^{30E/360}}$
  $\text{BPV} \leftarrow \text{BPV} + \delta_n^{30E/360} \cdot B(t_0, t_n^{\text{sw}})$
  $\mathcal{D} \leftarrow \mathcal{D} \cup \{(t_n^{\text{sw}},\, B(t_0, t_n^{\text{sw}}))\}$
end for
Interpolate $\mathcal{D}$ onto 16 quarterly schedule dates $\{t_1, \ldots, t_{16}\}$ (Following, Eurex DE)

Algorithm 2 Monte Carlo Coupon & Upfront Pricing

$r \leftarrow -\ln B(0, T)/T$ (continuous risk-free rate, $T = 4\text{y}$)
$\mathbf{S}^{(0)} \leftarrow (S_0^E,\, S_0^A)$ for all N_{sim} paths
for each daily step $j = 1, \ldots, 1016$ **do** ▷ 254 bdays/yr $\times$ 4 yr
  $k \leftarrow \lfloor (j-1)/64 \rfloor$ (quarterly index)
  Draw $\mathbf{Z}_j \sim \mathcal{N}(\mathbf{0}, \Sigma)$, $\Sigma_{12} = \rho = 0.4$
  $S_j^{(s)} \leftarrow S_{j-1}^{(s)} \cdot \exp\big[(r - q_s - \frac{\sigma_s^2}{2})\frac{1}{254} + \sigma_s \frac{1}{\sqrt{254}} Z_j^{(s)}\big]$, $s \in \{E, A\}$
  if $j \bmod 254 = 0$ **then** record annual monitoring price $\mathbf{S}^{(j/254)}$
  end if
end for
$R_i \leftarrow \frac{1}{2}\sqrt{\left(\frac{S_{t_n}^E}{S_{t_{n-1}}^E}\right)} + \frac{1}{2}\sqrt{\left(\frac{S_{t_n}^A}{S_{t_{n-1}}^A}\right)} - 1$ for path i
$PV_{\text{Coupon}} \leftarrow B(0, T) \cdot \alpha \cdot \dfrac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} \max(0, R_i)$
$PV_{\text{Euribor}} \leftarrow 1 - B(0, T)$
$PV_{\text{Spread}} \leftarrow s \cdot \sum_{i=1}^{16} B(0, t_i) \cdot \Delta t_i$
return $X \leftarrow PV_{\text{Euribor}} + PV_{\text{Spread}} - PV_{\text{Coupon}}$

Question 2 — Sensitivities

Algorithm 3 Equity Delta (Bump & Revalue)

$X_0 \leftarrow \textsc{Price}(S_0^E = 100,\ S_0^A = 200)$ (same seed)
$X_E \leftarrow \textsc{Price}(S_0^E = 101,\ S_0^A = 200)$
$X_A \leftarrow \textsc{Price}(S_0^E = 100,\ S_0^A = 201)$
$\Delta^E \leftarrow X_E - X_0, \quad \Delta^A \leftarrow X_A - X_0$

Algorithm 4 Parallel DV01 (Yield Curve Shift)

$X_0 \leftarrow \textsc{Price}(\mathcal{D})$
for each pillar $(t_i, B_i) \in \mathcal{D}$ **do**
    $r_i \leftarrow -\ln B_i / \tau_i$
    $\tilde{B}_i \leftarrow \exp\big[-(r_i + 10^{-4}) \cdot \tau_i\big]$
end for
$X_{\text{bump}} \leftarrow \textsc{Price}(\tilde{\mathcal{D}})$ (rerun full pricing with bumped curve)
$\text{DV01} \leftarrow X_{\text{bump}} - X_0$

Question 3 — Hedging

Algorithm 5 Delta and IR Hedge Construction

Equity hedge
$n^E \leftarrow -\Delta^E \cdot N, \quad n^A \leftarrow -\Delta^A \cdot N$ ▷ shares to buy/sell

IR hedge
$\text{DV01}^{1€}_{\text{IRS}} \leftarrow 10^{-4} \cdot \sum_{i=1}^{16} B(0, t_i) \cdot \Delta t_i$
$N_{\text{IRS}} \leftarrow N \cdot \dfrac{|\text{DV01}|}{\text{DV01}^{1€}_{\text{IRS}}}$
if $\text{DV01} > 0$ **then** enter *Receiver* IRS (pay Euribor, receive fixed)
else enter *Payer* IRS (pay fixed, receive Euribor)
end if